Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011) university under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: M.Tech(CSE)1st sem
Name of the Paper: Cryptography and Network Security
Paper Code: MCS-142
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks) (CO-1)

a. Given an example of sensitive information (a password), explain how cryptography would protect it during transmission.

OR

b. Describe the evolution of cryptographic methods from the classical ciphers to modern-day encryption algorithms.

Q2. (10 Marks) (CO-1)

a. How does asymmetric encryption differ from symmetric encryption, especially in terms of key management?

OR

b. Describe the difference between public key infrastructure (PKI) and symmetric key distribution in terms of security and usage.

Q3. (10 Marks) (CO-1)

a. Evaluate the effectiveness of key exchange protocols such as Diffie-Hellman in securely distributing keys over an untrusted network.

OR

b. How encrypt a message using RSA (asymmetric encryption) and explain how the encryption and decryption process works?

Q4. (10 Marks) (CO-2)

a. Apply the Blowfish encryption algorithm on a short message, and explain the steps involved in encryption.

OR

b. Given a message and a block cipher (e.g., AES), encrypt the message using ECB and CBC modes, and compare the results.

Q5. (10 Marks) (CO-2)

a. Given an elliptic curve and a public key, apply ECC encryption to a message and explain the decryption process.

OR

b. Using DSA or ECDSA, demonstrate how to create a digital signature for a message and verify it using the corresponding public key?